

Foto: Dr. Matthias Otte

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**EUROfusion**

**HELMHOLTZ**

RESEARCH FOR GRAND CHALLENGES



# IMPURITY TRANSPORT AND RADIATION AT THE STELLARATOR WENDELSTEIN 7-X

Promotionskolloquium

March 26, 2026

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# MOTIVATION

# NUCLEAR FUSION: ENERGY CHALLENGE

*Lawson criterion* for net energy gain:

$$n_e T \tau_E \geq \frac{12 f_{\text{tot}}}{\langle \sigma_{\text{DT}} v \rangle f_H^2 E_\alpha - 4 L_Z(T)} T^2$$
$$\gtrsim 3 \times 10^{21} \text{ keVs/m}^3$$

(requires temperatures  $T \sim 10^1$ - $10^3$  keV)

## Triple Product

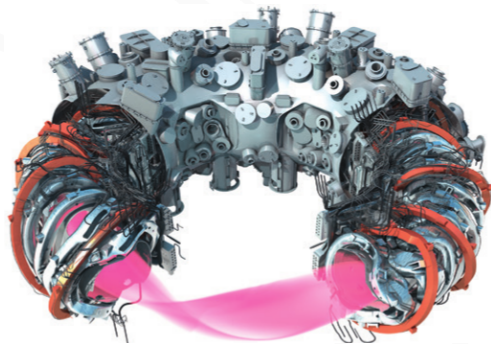
Key figure of merit for fusion performance combining density, temperature, and energy confinement time



# WENDELSTEIN 7-X

## Key Challenge

Managing heat loads on plasma-facing components during long, high performance plasma



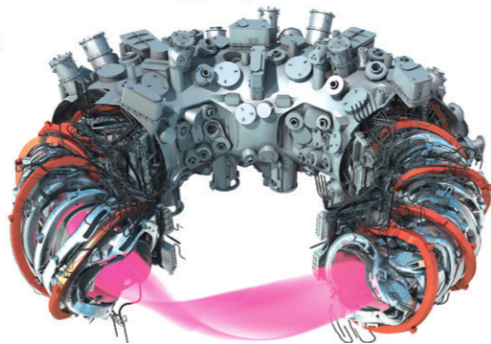
Source: © IPP

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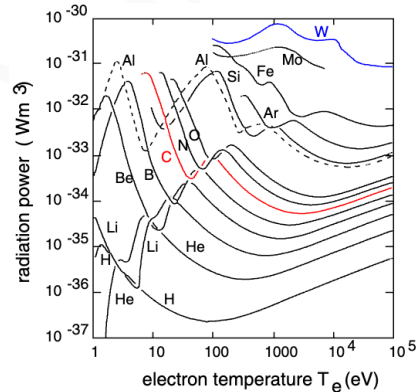
⇒ Dissipation of large fraction of heat load (energy) in scrape-off layer and particle flux to the target through detachment



Source: © IPP

# PLASMA RADIATION

- adjust plasma profiles so radiation in SOL increases and transport across separatrix reduced



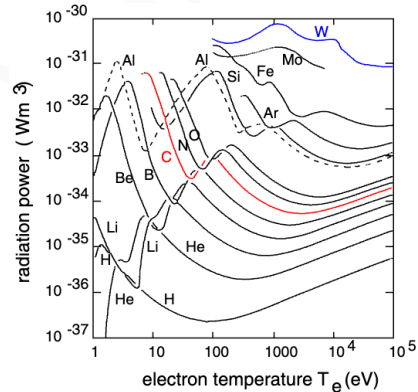
Source: R. Schneider et al., 2006[5]  
 $P_{\text{rad}} / (V n_e n_{\text{imp}})$  as a function of electron temperature

# PLASMA RADIATION

- adjust plasma profiles so radiation in SOL increases and transport across seperatrix reduced

## Impurity Seeding

Injection of low-Z (He, N<sub>2</sub>) or high-Z (Ne, Ar) impurities to enhance edge radiation and cooling



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 $P_{\text{rad}} / (V n_e n_{\text{imp}})$  as a function of electron temperature

# PLASMA RADIATION

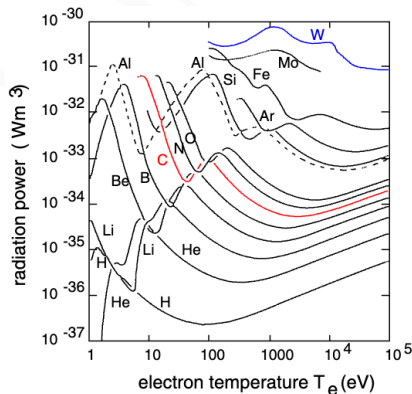
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## Impurity Seeding

Injection of low-Z (He, N<sub>2</sub>) or high-Z (Ne, Ar) impurities to enhance edge radiation and cooling

- avoid critical impurity accumulation, use working gas

⇒ stable control w. radiation information?



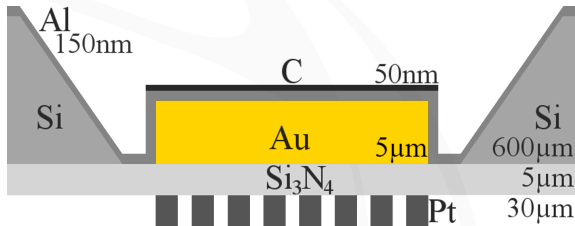
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 $P_{\text{rad}} / (V n_e n_{\text{imp}})$  as a function of electron temperature

# CORE BOLOMETER AT W7-X

# BOLOMETER DIAGNOSTIC

## Metal Resistor Bolometers:

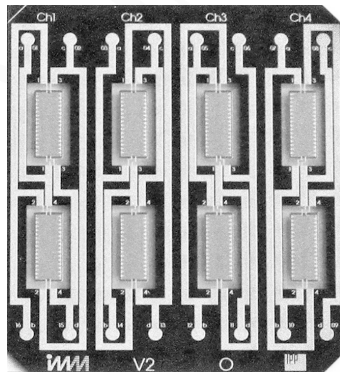
- Thin metal (gold) film absorbers
- Wheatstone bridge circuit configuration
- Multicamera system: HBC & VBCI/r (32 + 2×24 channels)
- Spatial resolution:  $\sim 5$  cm at magnetic axis



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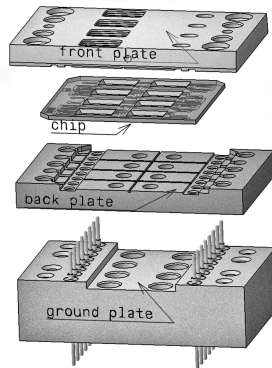
Source: D. Zhang et al., 2010[3]



# BOLOMETER DIAGNOSTIC

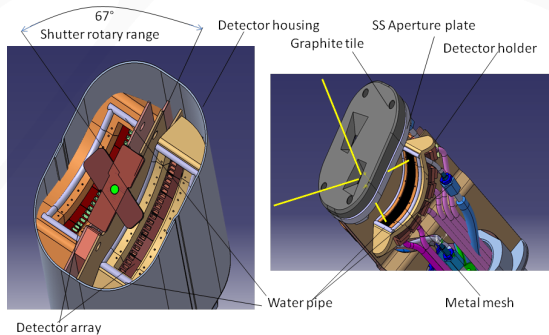
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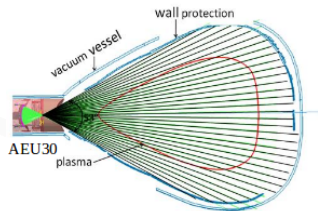


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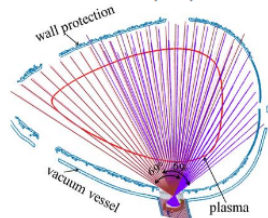
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Source: D. Zhang et al., 2010[3]



horizontal bolometer camera (HBC)

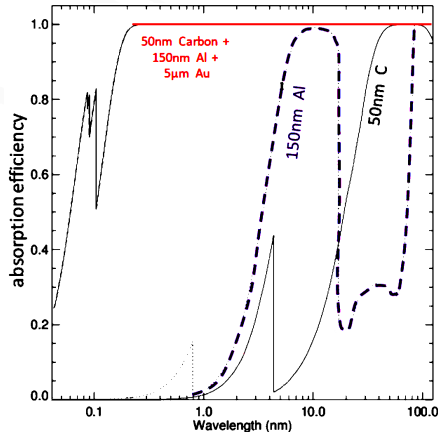


AEV21  
vertical bolometer camera (VBC)

# PERFORMANCE

## Performance Characteristics:

- Measures total radiation power (broadband)
- Near unity absorption efficiency 600-0.2 nm
- In-situ ohmic heating calibration with Wheatstone bridge ( $F_M$ ,  $f_M$ )
- $\{0.4, 0.8, \dots 6.4\}$  ms sampling
- *Gaussian* uncertainty:  
 $1.6 \mu\text{W} \leftrightarrow \text{SNR} > 1000$



Source: D. Zhang et al., ~2024[2]  
Relative absorption efficiency of absorber assembly

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## Power Per Detector:

$$P_M = F_M \cdot \left( \frac{d(\Delta \tilde{U}_M)}{dt} + f_M \Delta \tilde{U}_M \right)$$

## Global Radiation Power:

$$P_{\text{rad}} = \frac{V_{\text{P,tor}}}{V_C} \sum_M \frac{P_M V_M}{K_M}$$

# PERFORMANCE

## Chord Brightness

Line-integrated measurement along each detector's line of sight provides radial profile information ( $P_{\text{ch}} \sim P_M/K_M$ )

## Applications:

- Power balance calculations
- **Tomographic reconstruction**
- **Real-time feedback control**

⇒ controlled cooling in the plasma edge

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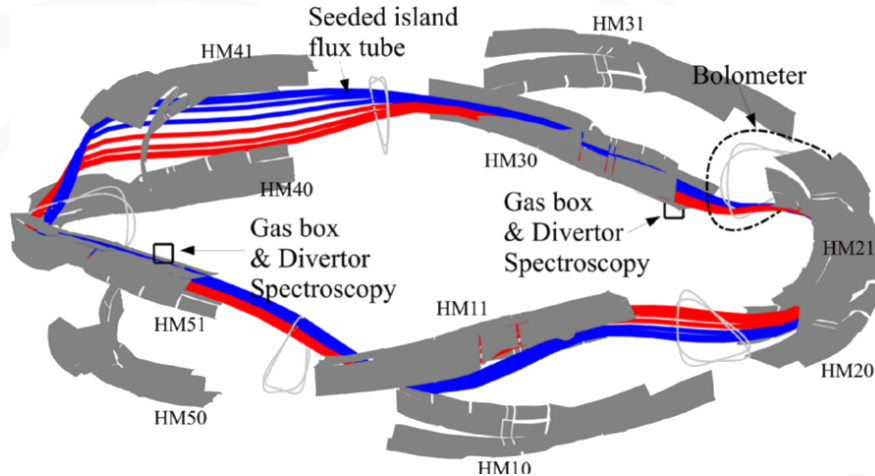
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# REAL-TIME RADIATION FEED- BACK CONTROL

# FEEDBACK SYSTEM



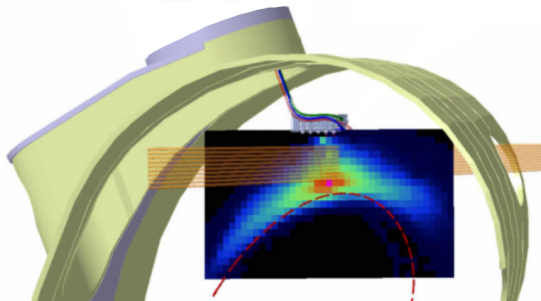
Source: F. Effenber et al., 2019[4]

Divertor elements HM 1-5: up (1) down (0) & gas boxes, bolometry system and connecting field lines

# FEEDBACK SYSTEM

## Architecture:

- Integrated data acquisition hardware & optimized control software
- Twin proxies and dedicated signal outputs
- Thermal gas valve actuation with multi-diagnostic PID controller



Source: T. Barbui et al., 2016[1]  
Thermal helium beam and spectroscopy



# FEEDBACK SYSTEM

## Architecture:

- Integrated data acquisition hardware & optimized control software
- Twin proxies and dedicated signal outputs
- Thermal gas valve actuation with multi-diagnostic PID controller

## Radiation Prediction Proxies:

**A:** *Channel selection*  $S$

$$P_{\text{pred}}^{(1)} = \frac{V_{P,\text{tor}}}{V_S} \sum_M^S \frac{P_M V_M}{K_M}$$

(e.g.  $S_{\text{VBC}} = \{61, 86, 78, 73, 70, 65\}$ )

**B:** *Single channel*

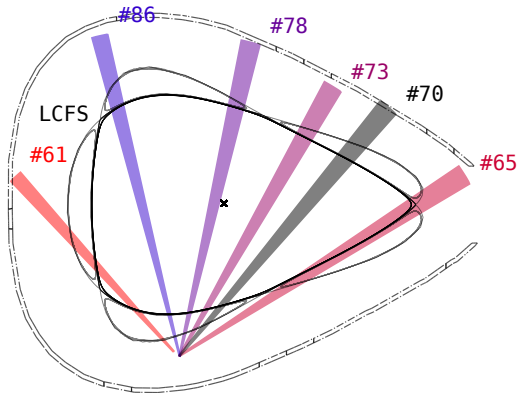
$$P_{\text{pred}}^{(2)} \propto \Delta \widetilde{U}_M$$

(dimensionless, minimal overhead)

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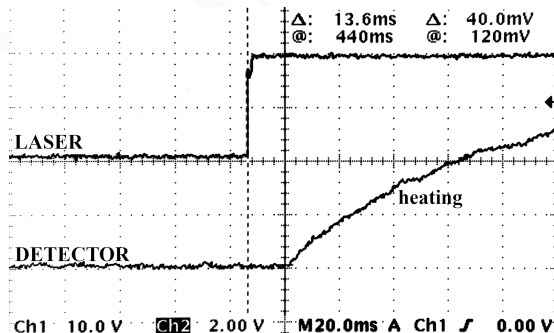


Line of sight selection  $S$  (XP20181010.32), only VBC channels covering magnetic islands, as well as the core and X-points partially

# FEEDBACK SYSTEM

## Performance:

- FIFO smoothing in  $P_{\text{pred}}^{(1)}$ :  $\sim N\Delta t/2$
- Minimum acquisition latency 13.6 ms
- Total system latency: 150-400 ms (dominated by plasma response)



oscilloscope screen capture; single detector  
laboratory latency measurement with laser diode

# EXPERIMENTAL ACHIEVEMENTS

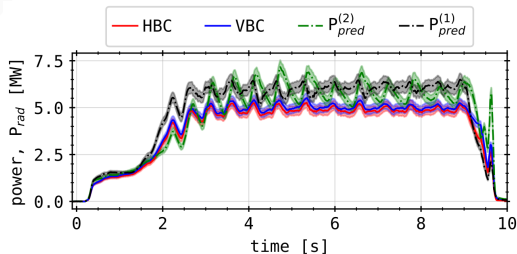
## XP20181010.32

- Radiation fraction PID setpoint  
 $f_{\text{rad}} \sim 90\% \text{ off } P_{\text{pred}}^{(1)}$
- 5-channel VBC subset  
 $S = \{61, 65, 73, 78, 86\}$

# EXPERIMENTAL ACHIEVEMENTS

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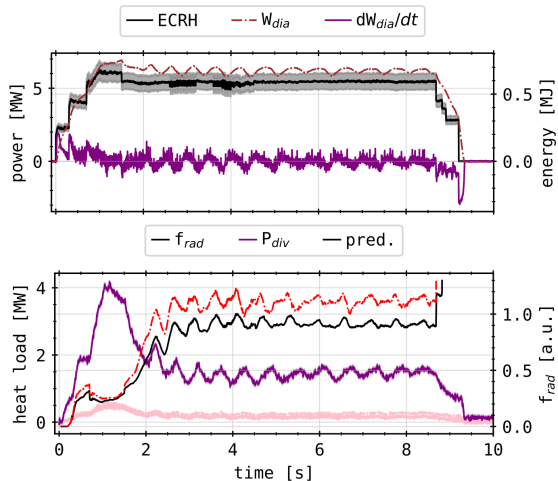
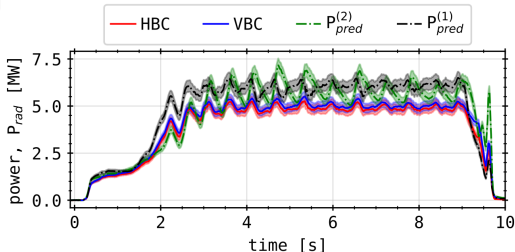
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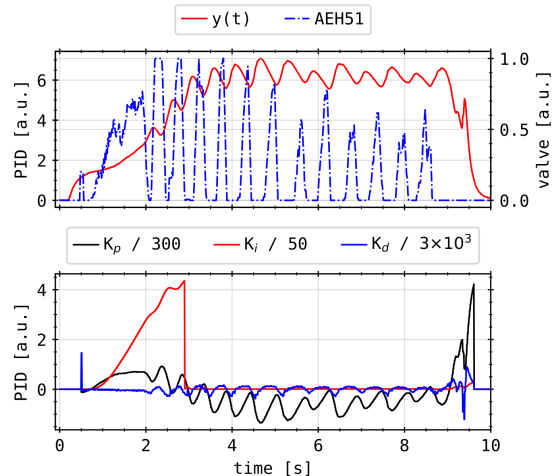
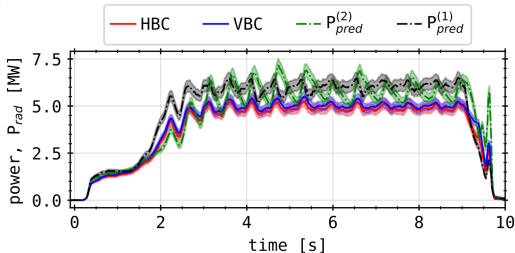
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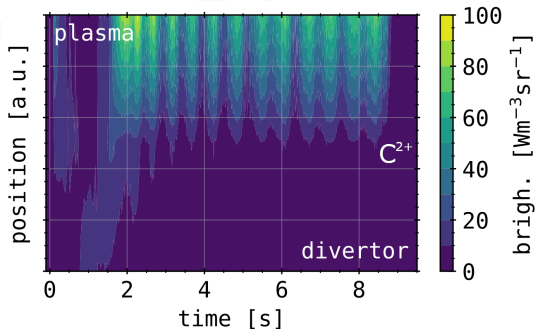
# EXPERIMENTAL ACHIEVEMENTS

## XP20181010.32

- Peak  $f_{\text{rad}} \gtrsim 100\%$ , cycling with gas inlet
- Target heat load reduction factor  $\geq 2$
- $\text{C}^{2+}$  detachment visible at  $f_{\text{rad}} \sim 50\%$

### Key Result

Demonstrated stable, real-time feedback-controlled radiative cooling with  $f_{\text{rad}} \geq 85\%$  without terminal plasma disruption and significant target heat load reduction



spatial-temporal evolution ( $\text{C}^{2+}$  radiation)  
in divertor of HM51; filterscope measurements



# LINE-OF-SIGHT SENSITIVITY

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- Evaluate LOS/channel efficacy for feedback (generalised local radiation sensitivity)
- Data set of all feedback discharges from OP1.2b
- Applied 5 different metric (linear & weighted deviation, FFT, ...)

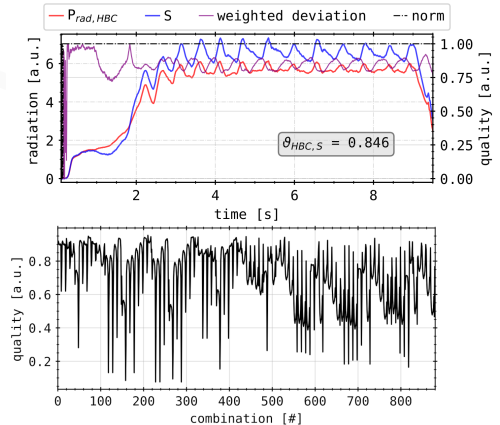
## Weighted deviation:

$$\varphi(t) = 1 - \frac{\|P_{\text{pred}} - P_{\text{rad}}\|(t)}{P_{\text{rad}}(t)}$$

$$\vartheta = \frac{1}{T} \int_T \varphi(t) dt$$

# LINE-OF-SIGHT SENSITIVITY

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- Data set of all feedback discharges from OP1.2b
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(top) weighted deviation norm and example for  $S_{HBC} = \{8, 7, 23\}$ ;  
 (bottom) full dataset for  $M = 3$  selection sets (HBC)

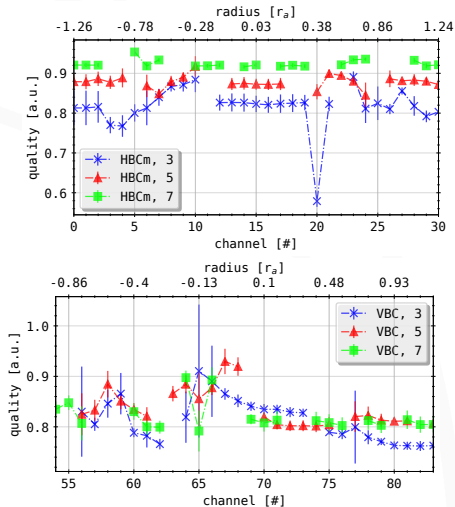
# LINE-OF-SIGHT SENSITIVITY

## Optimal Channel Selection:

- No single "best" set exists
- Robust selection achieves  $\geq 85\%$  prediction accuracy
- VBC channels generally outperform HBC for edge sensitivity
- 3-5 channels sufficient for reliable  $P_{\text{pred}}$

## Takeaway

Detectors viewing close to separatrix and SOL most viable for feedback purposes



LOS sensitivity by weighted deviation measurement for  $M = 3, 5, 7$

# TOMOGRAPHY

# MINIMUM FISHER REGULARIZATION

Measurement:

$$\mathbf{T} \vec{x} = \vec{b}$$

**Bolometer tomography:**

- 61 line-integrated measurements  $\vec{b}$
- 2D emissivity  $\vec{x}$ :  $\sim 4500$  pixels
- *Geometry*  $\mathbf{T}$  has high condition

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- 2D emissivity  $\vec{x}$ :  $\sim 4500$  pixels
- *Geometry*  $\mathbf{T}$  has high condition
- ill-posed, requires regularization

Problem:

$$\min_{\eta} \left\| \left( \mathbf{T}^T \mathbf{T} + \mu^{-1} \mathbf{H} \right)^{-1} \mathbf{T}^T \vec{b} \right\| \rightarrow \vec{x}$$

Fisher Information:

$$I_F = \int \frac{1}{x} \left( \frac{\partial x}{\partial \vec{r}} \right)^2 d\vec{r} \leftrightarrow \mathbf{H} \sim \frac{1}{x} \Delta^T \Delta$$

limiting error on reconstruction:

$$\sigma_x \leq \frac{1}{I_F}$$

# MINIMUM FISHER REGULARIZATION

## Radially Dependent Anisotropy:

- Tailored weighting to poloidal gradient
- $k_{\text{ani}} < 1$ : favours anisotropy
- $k_{\text{ani}} > 1$ : favours smooth emissivity

$$\mathbf{H} \rightarrow \mathbf{H}_{\text{ani}}$$

$$\Delta = \nabla_r^T(\cdot) \nabla_r + \nabla_\vartheta^T(\cdot) \nabla_\vartheta$$

$$\nabla_\vartheta \rightarrow \tilde{\nabla}_\vartheta = \frac{1}{k_{\text{ani}} \Delta r} \mathbf{D}_{\text{dia}}^{(0)} \nabla_\vartheta$$

## MFR + RDA Advantages

- Robust to noisy data
- Incorporates a priori knowledge
- Stable convergence



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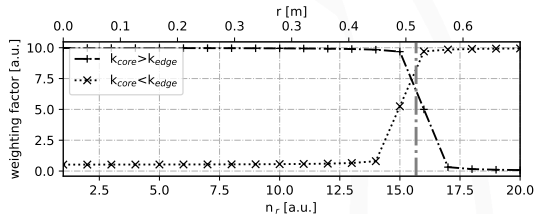
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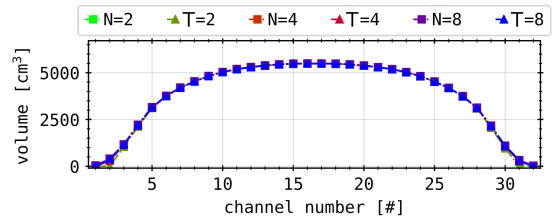
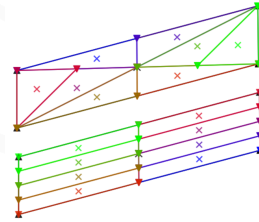


$k_{\text{ani}}$  profile examples over radial cell number

# BENCHMARKING

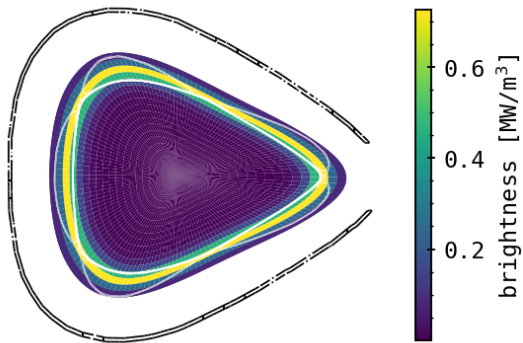
## Geometry Perturbation Test:

- Previous and future calculations based on as-designed geometry



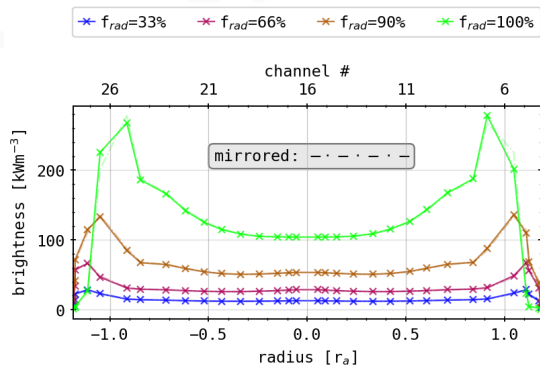
(top) detector/aperture splitting  $N = 8$   
 (bottom)  $V_M$  comparison for different segmentations

# BENCHMARKING



EIM grid, STRAHL phantom radiation profile

( $f_{\text{rad}} \sim 100\%$ )



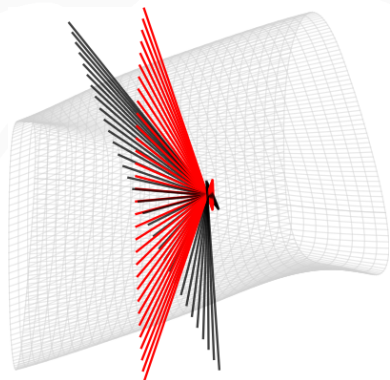
forward integration of STRAHL phantom profile  $f_{\text{rad}} \sim 100\%$

# BENCHMARKING

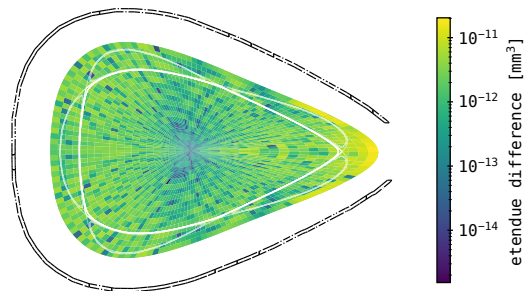
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- Tests for symmetrical chord profile yield artificial camera geometry

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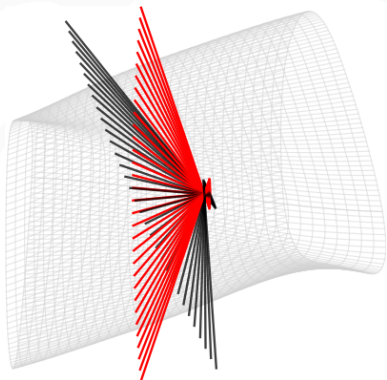
LOS comparison: original vs. artificial, symmetric HBC



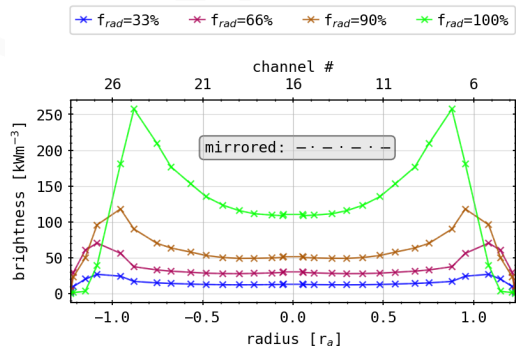
absolute difference between camera geometries (left, original vs. symmetric)

$$\left\| \mathbf{T} - \mathbf{T}_{\text{sym}} \right\|$$

# BENCHMARKING



LOS comparison: original vs. artificial, symmetric HBC



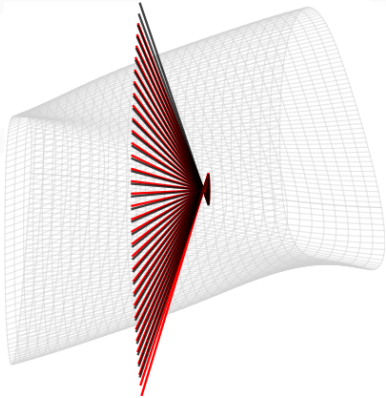
forward integration of STRAHL phantom profile  
for  $f_{rad} \sim 100\%$  with symmetrical camera geometry

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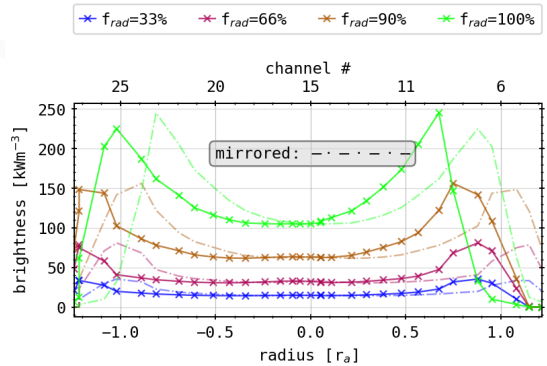
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- Perturbation study from measured machine stress  $\leq \pm 2^\circ$  off axis

# BENCHMARKING



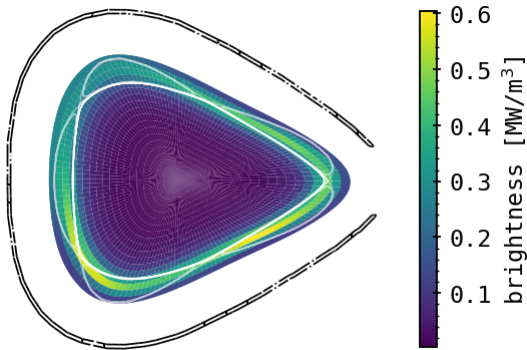
tilted (symmetrical) camera geometry;  $2^\circ$  poloidal rotation at aperture



forward integration of STRAHL phantom profile  
using  $2^\circ$  rotated symmetrical geometry

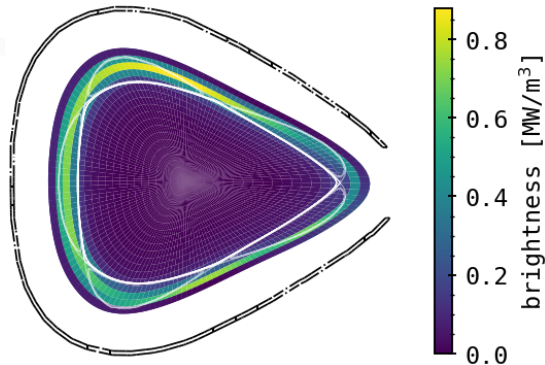


# BENCHMARKING



tomogram of forward calculation using tilted geometry, backward symmetrical

$$\text{using } \mathbf{T}_{\text{sym}} \cdot \vec{x} = \vec{b}$$



tomogram of forward and backward calculation using symmetrical geometry

$$\text{using } \mathbf{T}_{\text{ilt}} \cdot \vec{x} = \vec{b}$$

# BENCHMARKING

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## Results

- Discretization of system is computationally stable
- Intrinsic asymmetry of same order of magnitude as potential, unknown geometric perturbation - non-negligible
- Worst case machine warp scenario  $\sim 0.2r_a$  shift

# PHANTOM RECONSTRUCTIONS

## Purpose

Regularized, parametric inversions need experienced tailoring for their respective application.

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- 12+ synthetic emissivities: Gaussian, rings, anisotropies, convolutions...
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- Performed for only one magnetic grid configuration & LOS geometry

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## Metrics:

$$\rho_c = \frac{1}{\sigma_{pha}\sigma_{tom}} \text{cov}(\vec{x}_{\text{phan}}, \vec{x}_{\text{tom}})$$

$$M_{\text{SD}} \sim \left\| \frac{\vec{x}_{\text{tom}} - \vec{x}_{\text{phan}}}{\|\vec{x}_{\text{phan}}\|} \cdot \vec{A}^T \right\|$$

$$P_{\text{rad},X}^{2\text{D}} = \vec{x}_X^T \cdot \vec{A}, \quad X = \{\text{phan}, \text{tom}\}$$

$$\chi^2 = \frac{1}{N_{\text{ch}}} \left\| (\vec{b}_{\text{phan}} - \vec{b}_{\text{tom}}) \vec{\sigma}_{\text{bol}}^{-1} \right\|$$

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Regularized, parametric inversions need experienced tailoring for their respective application.

## Benchmarks:

- 12+ synthetic emissivities: Gaussian, rings, anisotropies, convolutions...
- $k_{\text{ani}}$  & phantom variations studies (radius, intensity etc.)
- Performed for only one magnetic grid configuration & LOS geometry

## Metrics:

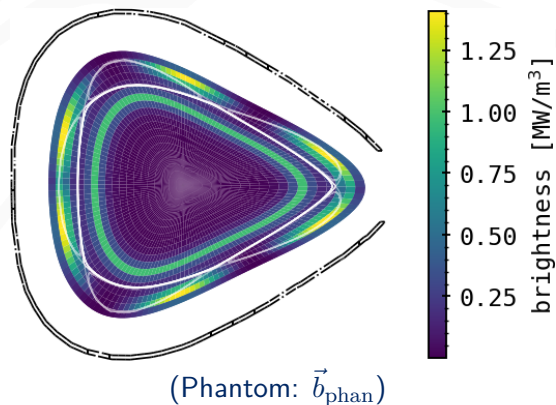
$$\rho_c = \frac{1}{\sigma_{pha}\sigma_{tom}} \text{cov}(\vec{x}_{\text{phan}}, \vec{x}_{\text{tom}})$$

$$M_{\text{SD}} \sim \left\| \frac{\vec{x}_{\text{tom}} - \vec{x}_{\text{phan}}}{\|\vec{x}_{\text{phan}}\|} \cdot \vec{A}^T \right\|$$

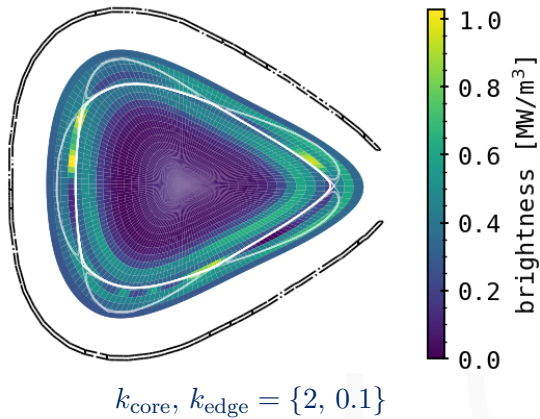
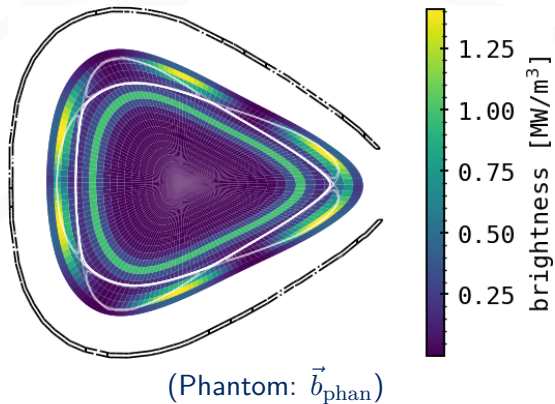
$$P_{\text{rad},X}^{2\text{D}} = \vec{x}_X^T \cdot \vec{A}, \quad X = \{\text{phan}, \text{tom}\}$$

$$\chi^2 = \frac{1}{N_{\text{ch}}} \left\| \left( \vec{b}_{\text{phan}} - \vec{b}_{\text{tom}} \right) \vec{\sigma}_{\text{bol}}^{-1} \right\|^2$$

# PHANTOMS: EXAMPLE

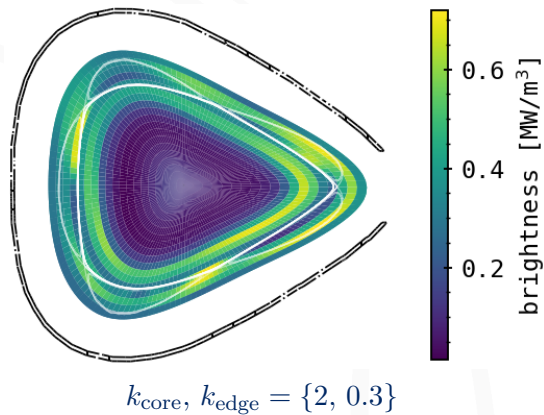
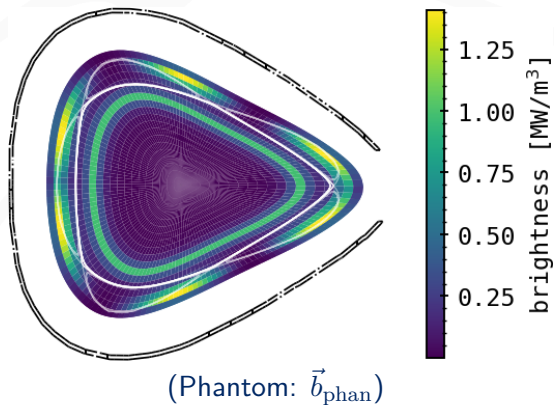


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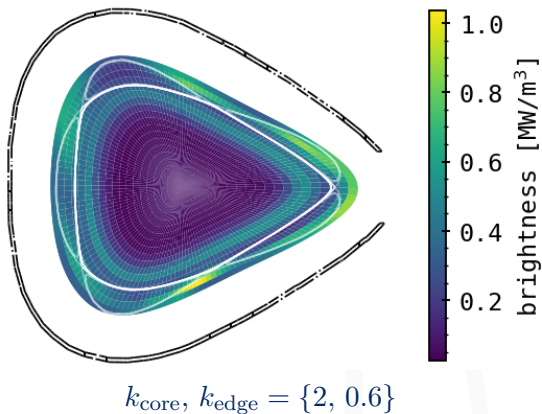
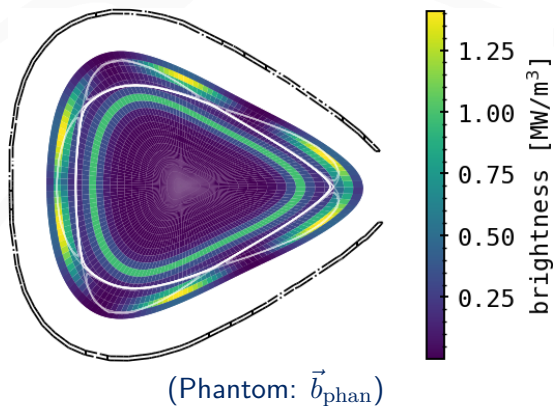




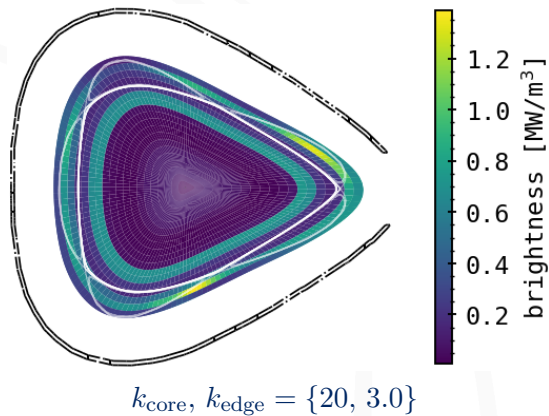
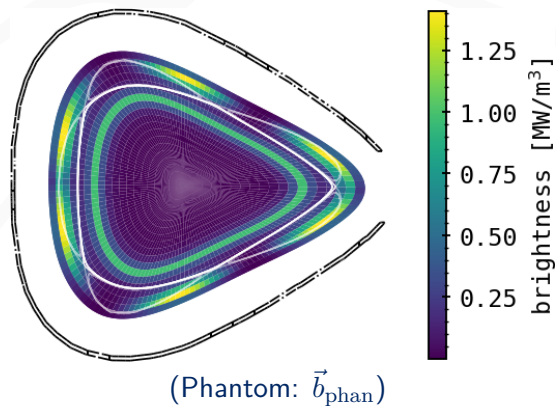
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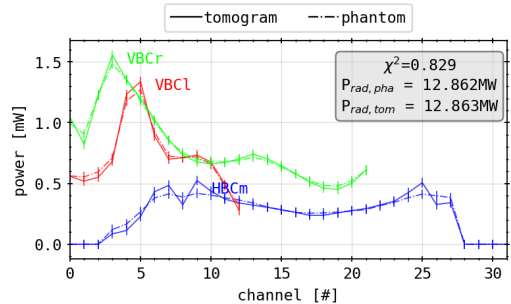
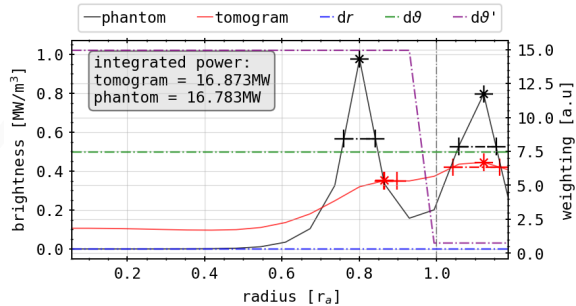
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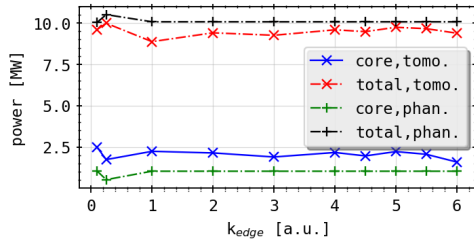
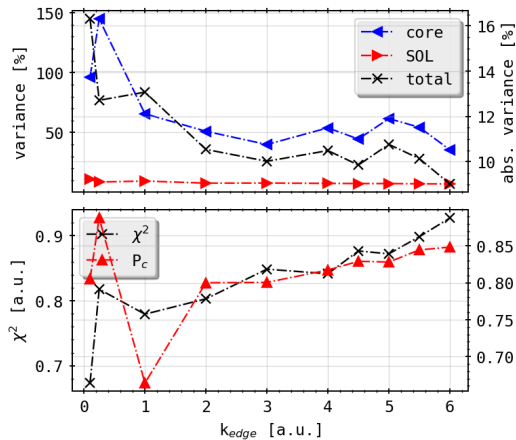
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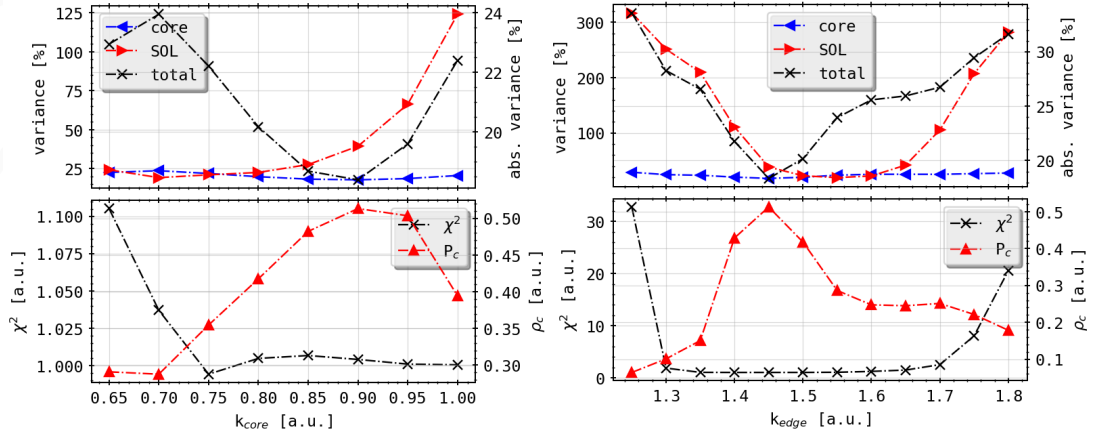


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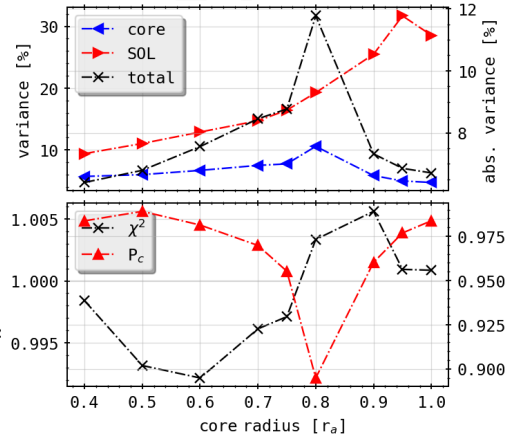
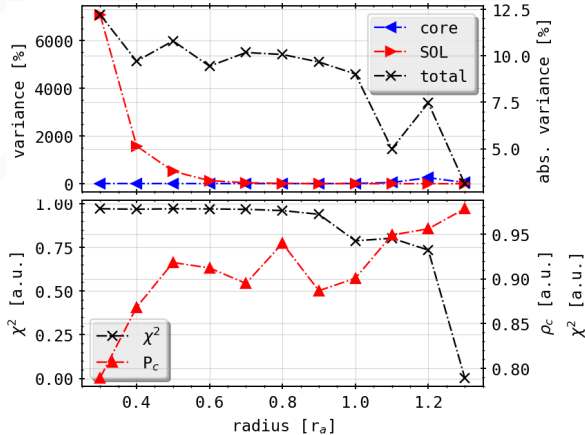
# PHANTOMS: STATISTICS

## Isotropic Rings: $k_{\text{ani}}$ scan



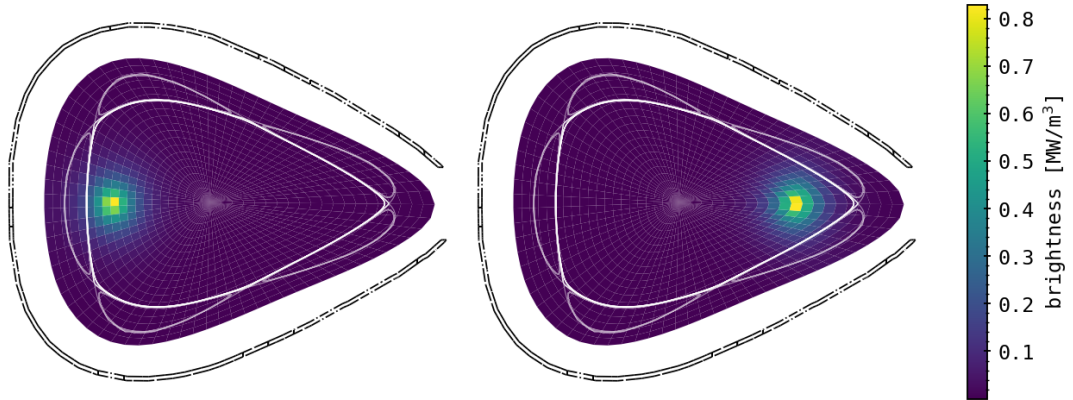
# PHANTOMS: STATISTICS

## Isotropic Rings: Radial Scan



# PHANTOMS: STATISTICS

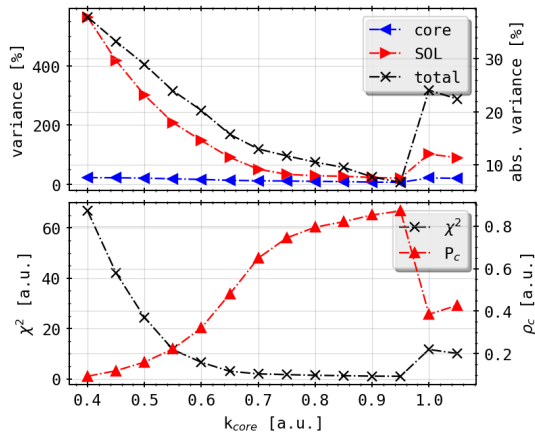
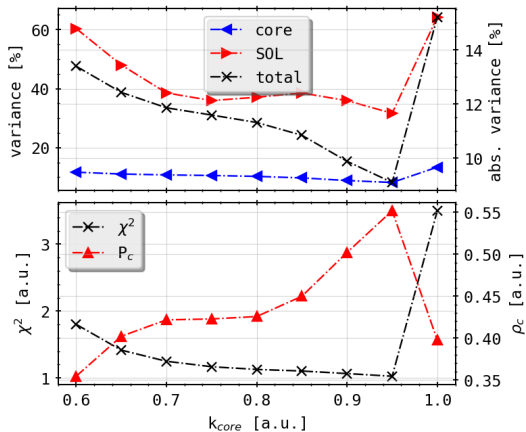
## Composite Distributions





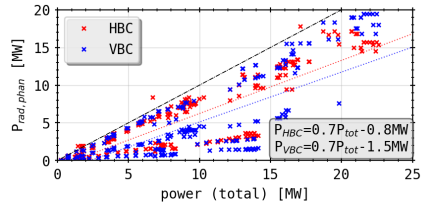
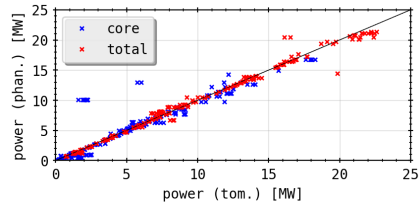
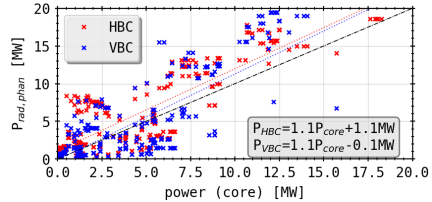
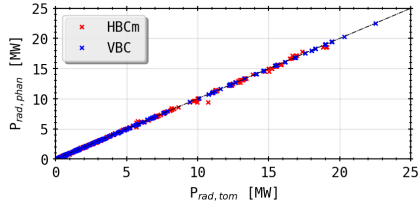
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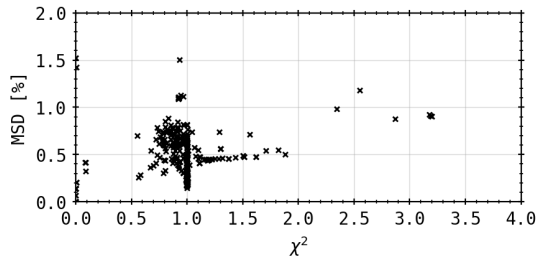
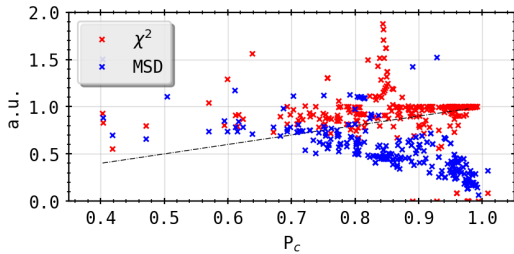
# PHANTOMS: STATISTICS

## Full Set



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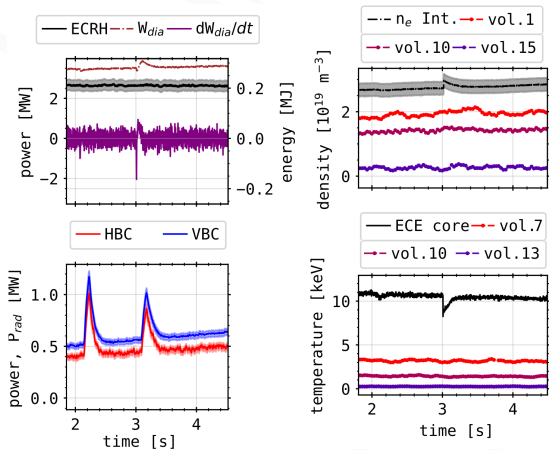
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## Conclusion

MFR with RDA weighting provides robust, reliable 2D radiation reconstructions suitable for further analysis and calculations.

# EXPERIMENTAL RECONSTRUCTIONS

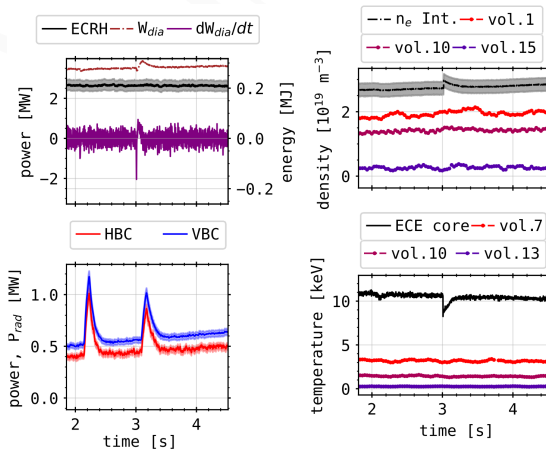
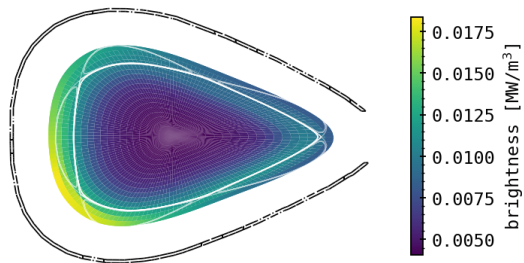
Example: XP20180809.13





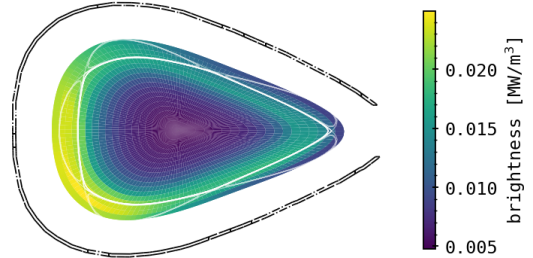
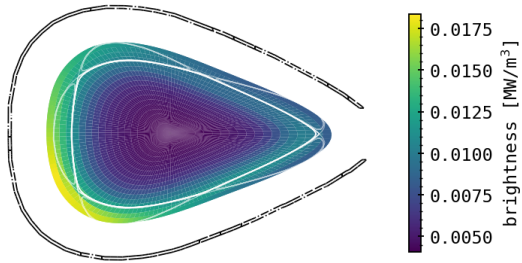
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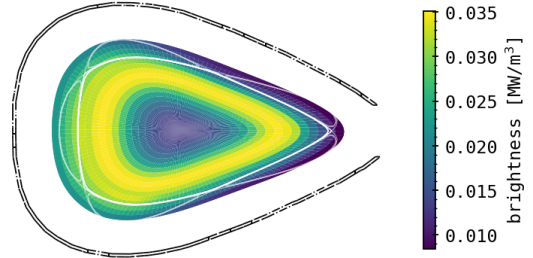
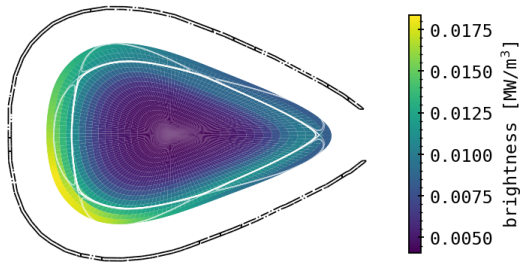
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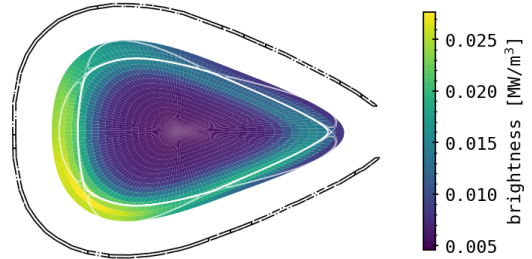
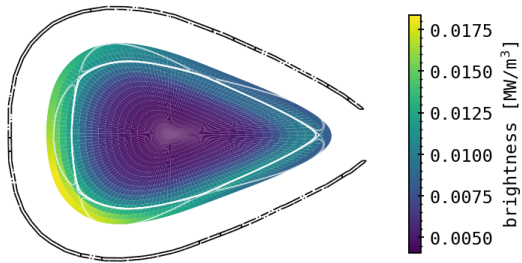
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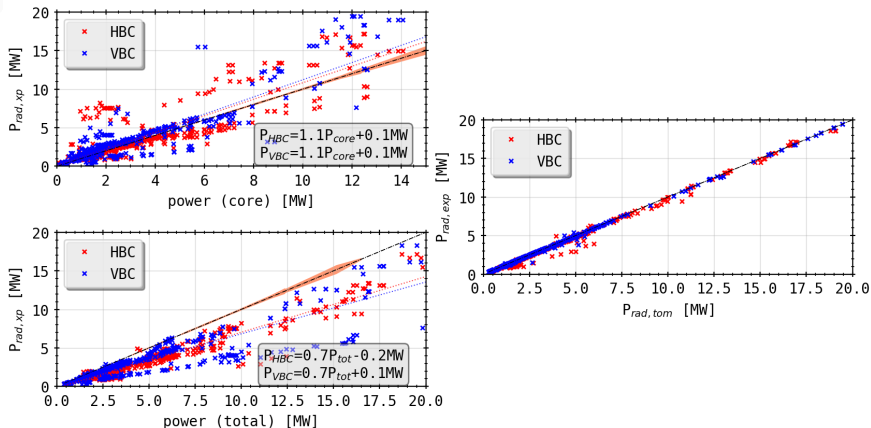
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Example: XP20180809.13



# EXPERIMENTAL RECONSTRUCTIONS

## Statistics: Full Set



# GLOBAL POWER BALANCE

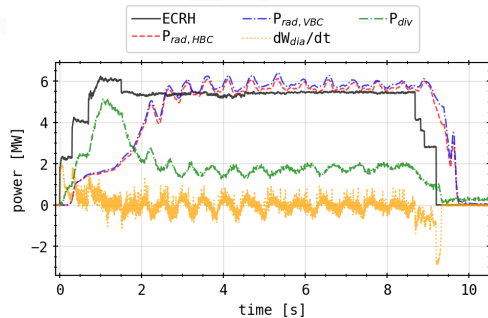
**Simplified 0D Ansatz:**

$$\begin{aligned}
 0 &\stackrel{!}{=} S + \frac{1}{V_P} \int \frac{3}{2} \left( \frac{\partial p}{\partial t} \right) + p \nabla \vec{v} + \nabla \vec{q} d\vec{r} \\
 &= -P_H + P_{\text{rad}} + P_{\text{div}} + \frac{dW}{dt} \\
 &= P_{\text{bal}}
 \end{aligned}$$

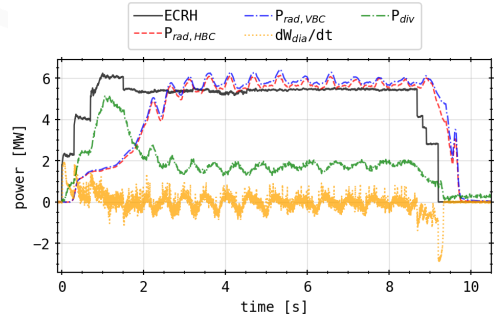
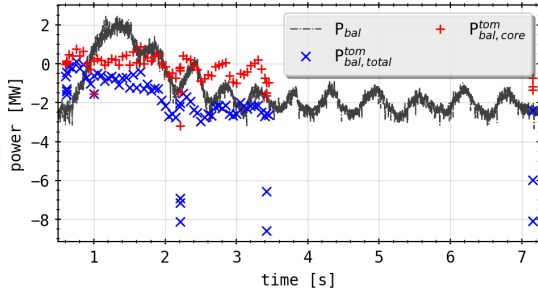
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# GLOBAL POWER BALANCE





# EXPERIMENTAL RECONSTRUCTIONS

## Statistical Analysis

- $P_{\text{rad}} \propto P_{2\text{D}}^{(X)}$  correlation (core, SOL)
- 2D integrated total power significantly underestimated (conflict experimental experience)
- SOL emissivity over-emphasized in reconstruction and absolute power figures
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## Power Balance

- $P_{2\text{D}}$  improves balance accuracy and stability
- Strong improvement over  $P_{\text{rad}}$  in transient scenarios
- Integrated 2D core emissivity nearly satisfies  $P_{\text{bal}} = 0$
- Viable alternative for quasi steady-state analysis

# CONCLUSIONS

# KEY ACHIEVEMENTS

## 1. **Real-Time Radiation Feedback System:**

Implemented, tested & operated first bolometer radiation feedback at W7-X

## 2. **Line-of-Sight Optimization:**

Systematic sensitivity analysis for optimal detector configurations

## 3. **Tomographic Reconstruction:**

Benchmarked and tailored MFR algorithm with *RDA* weighting

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- First controlled ( $P_{\text{rad}}$ ), stable detachment with  $f_{\text{rad}} \geq 85\%$
- Target heat load reduction factor  $\geq 2$
- Validated 3-5 channel LOS subsets for fast prediction ( $\geq 80\%$  accuracy)

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- Separatrix and SOL viewing channels most effective
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## 3. Tomographic Reconstruction:

Benchmarked and tailored MFR algorithm with *RDA* weighting

- Extensive artificial emissivity profile testing & parameter variation
- Reconstruction quality  $\geq 80\%$  for well-constrained phantoms
- Successful Application to experimental data & power balance calculations

# OUTLOOK



# FUTURE AND RECOMMENDATIONS

## System Upgrades:

- Reduce feedback latency  $< 10$  ms
- Implement machine learning for  $P_{\text{pred}}$ /LOS selection
- Per-experiment configuration ( $K_M$ ,  $V_M$ )
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## Tomography Extensions:

- Larger phantom benchmark set ( $> 50$  cases) and parameter space
- Real-time algorithms (benchmark LUT?)
- Automated parameter optimization (grid search, ML)

# COLLABORATIONS

## Acknowledgement

This work has been carried out within the framework of the EUROfusion Consortium and has received funding from the Euratom research and training programme 2014-2018 under grant agreement No. 633053. The views and opinions expressed herein do not necessarily reflect those of the European Commission.

## Collaborators:



**HELMHOLTZ**  
RESEARCH FOR GRAND CHALLENGES



**EUROfusion**



# Thank You

# REFERENCES I

- [1] T. Barbui et al. "Feasibility of line-ratio spectroscopy on helium and neon as edge diagnostic tool for Wendelstein 7-X". In: *Review of Scientific Instruments* 87.11 (2016), 11E554. DOI: [10.1063/1.4962989](https://doi.org/10.1063/1.4962989). eprint: <https://aip.scitation.org/doi/pdf/10.1063/1.4962989>. URL: <https://aip.scitation.org/doi/abs/10.1063/1.4962989>.
- [2] "Calculations for detection efficiency from tabulated coefficients of Henke et al [Henke B.L., Gullikson E.M. and Davis J.C. 1993 At. Data Nucl. Data 54 181]". In: (2024).
- [3] D. D. Zhang et al. "Design criteria of the bolometer diagnostic for steady-state operation of the W7-X stellarator". In: *Review of Scientific Instrument* 81.10E134 (2010).

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- [4] F. Effenberg et al. "First demonstration of radiative power exhaust with impurity seeding in the island divertor at Wendelstein 7-X". In: *Nuclear Fusion* 59.10 (2019), p. 106020. DOI: [10.1088/1741-4326/ab32c4](https://doi.org/10.1088/1741-4326/ab32c4). URL: <https://doi.org/10.1088/1741-4326/ab32c4>.
- [5] R. Schneider et al. "Plasma Edge Physics with B2-Eirene". In: *Contributions to Plasma Physics* 46.1-2 (2006), pp. 3–191. DOI: <https://doi.org/10.1002/ctpp.200610001>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/ctpp.200610001>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/ctpp.200610001>.